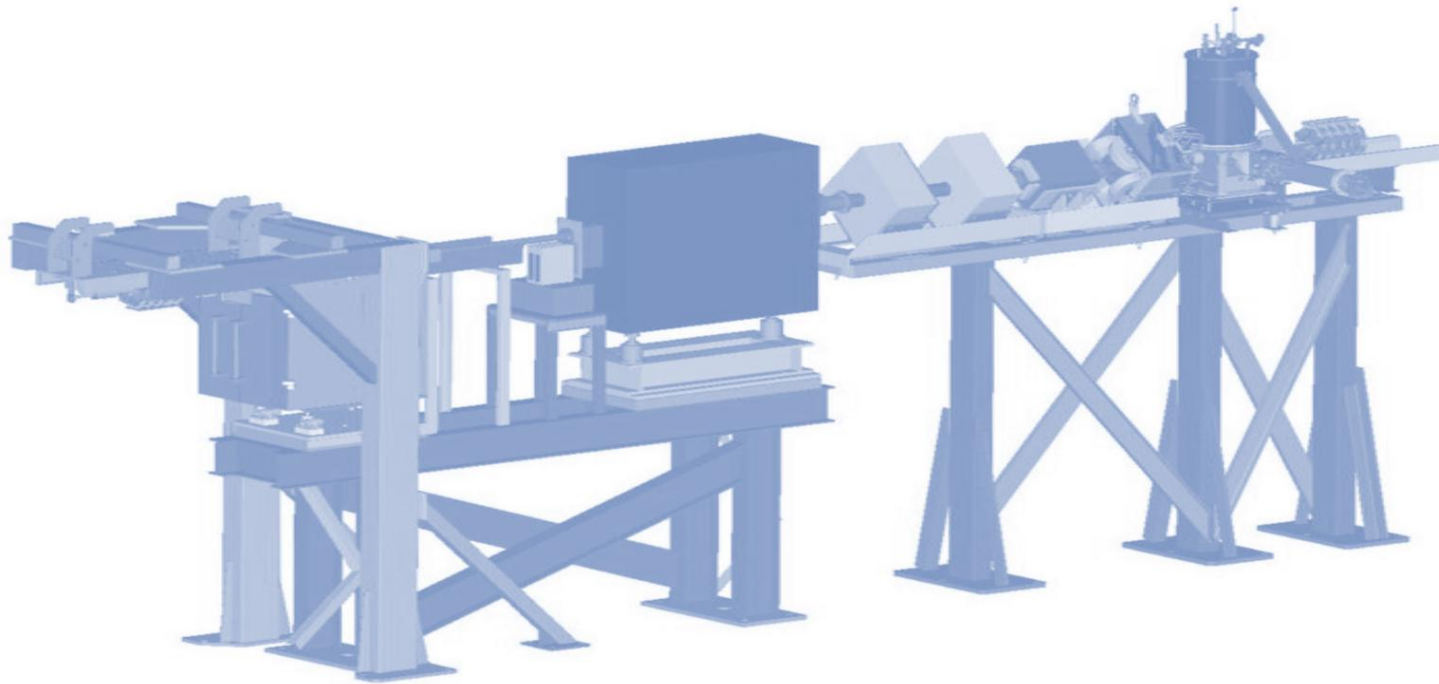


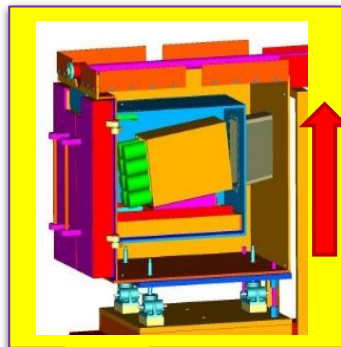
HALL A MØLLER POLARIMETER STATUS



O. Glamazdin, R. Pomatsalyuk, V. Vereshchaka
National Science Center
“Kharkov Institute of Physics and Technology”

J. Gomez
Jefferson Lab

11 GeV Upgrade



$10^\circ \rightarrow 7.3^\circ$



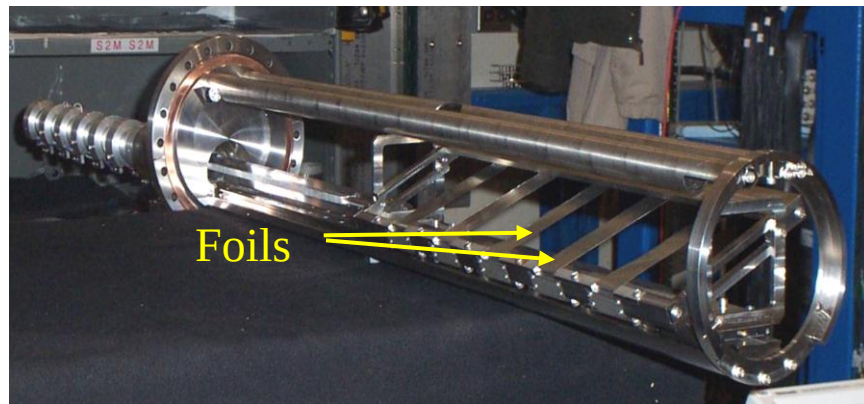
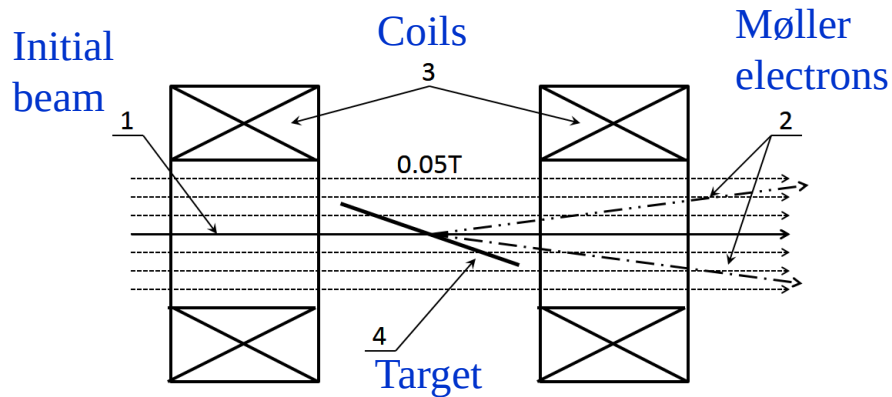
- Shielding pipe in dipole
- New detector shield box
- Reconfigured girder:
v. corrector + quad + BPM

- 1st quad shifted downstream
- New 4th quad added
- New girder for quads #1 and #4

TWO TYPE OF POLARIZED ELECTRON TARGETS

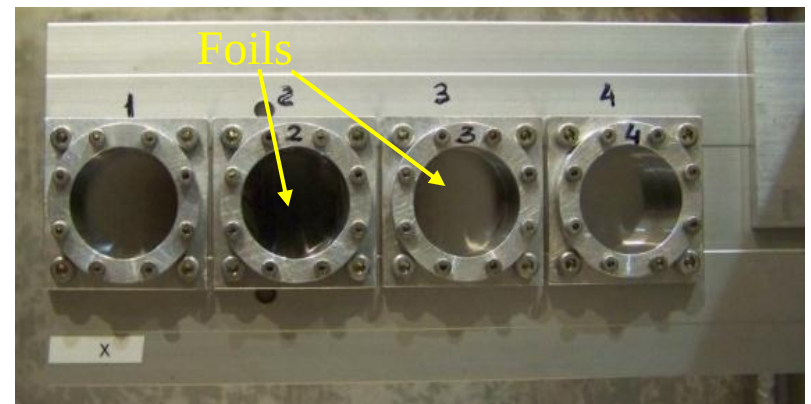
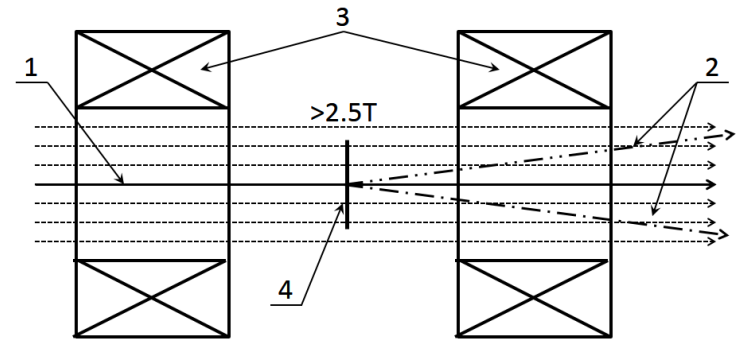
“Low Field” (operating since 2005):

- Longitudinally polarized ferromagnetic foils
- 7-30 μm thick (tilted at 20° to the beam)
- Low magnetic field ~ 250 Gs
- Sys. Errors $\sim 1.7\%$



“High Field” (operating since 2010):

- Transverse polarized ferromagnetic foils
- 1-12 μm thick (set at 90° to the beam)
- High magnetic field ~ 4 T
- Sys. Errors $\sim 0.9\%$



COMMISSIONING RUN

April 02 – 03, 2014

The beam energy was 6054 MeV - 6113 MeV

The beam current 0.2 - 0.5 μ A

Wien filter angle was 0.0°.

IHWP wave plate was OUT. Passive HWP was OUT.

Raster was OFF.

Helicity frequency=30Hz (1kHz), delay=8windows, pattern=Quartet.

Both old DAQ and FADC DAQ were running in parallel.

18:48 - tune beam in the Hall.

19:12 - CW beam in the Hall.

Moller settings took ~1 hour (20:20 - 21:20) (Thanks Yves).

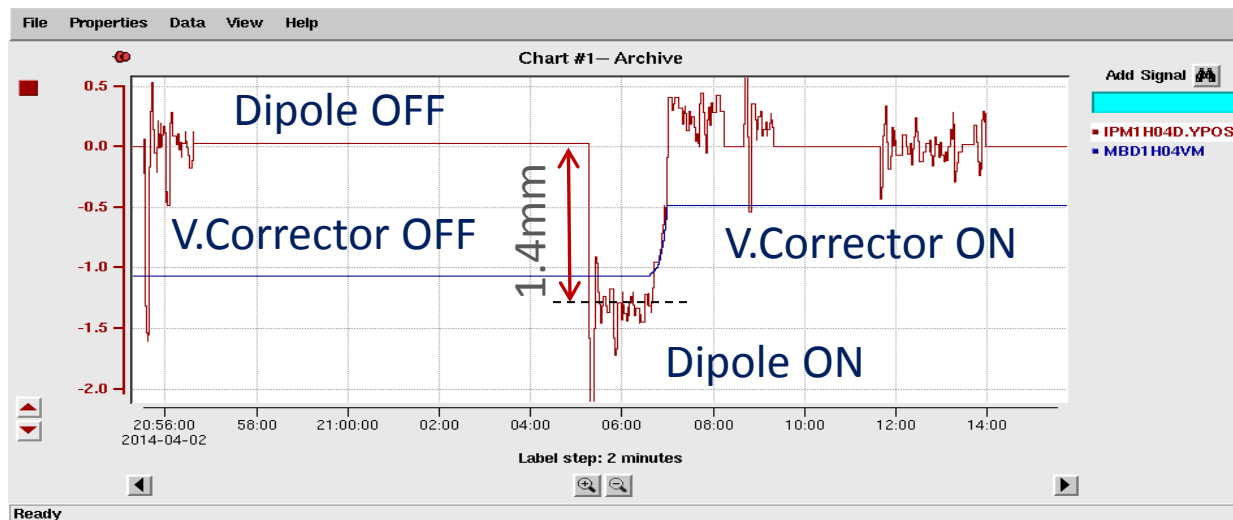
21:23-22:05 - detector HV tune.

22:10-1:43 - Beam polarization measurement with targets 5,4,3,2,Al.

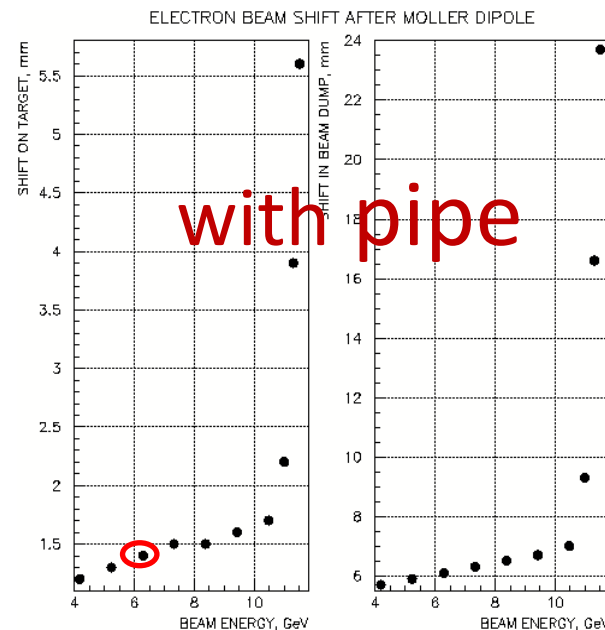
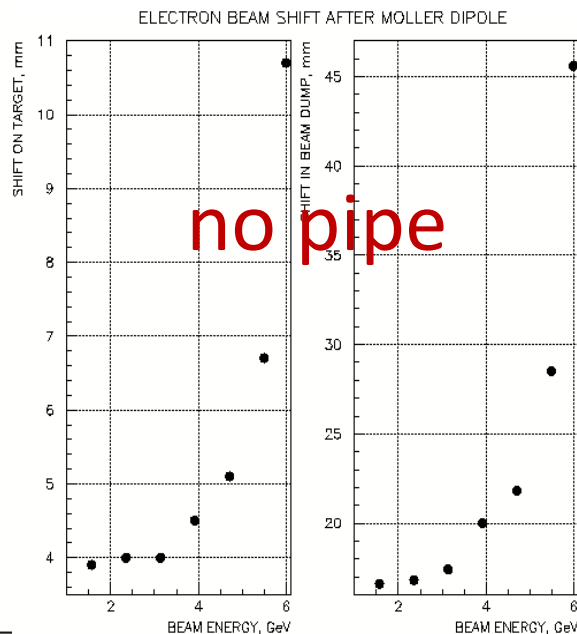
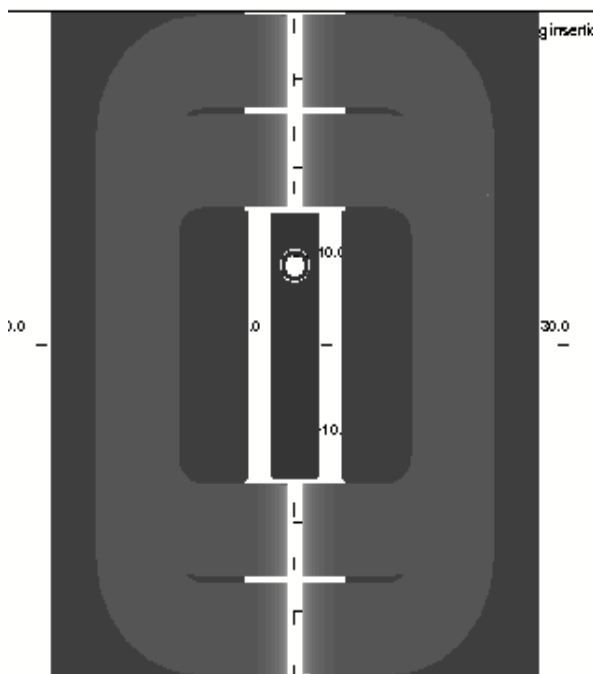
Measured asymmetry with **Al** target (non-polarized) 0.0004 $\pm$ 0.0007

1:46–3:00 – DAQs and Analyzers check with helicity 30Hz, and 1kHz

TEST of NEW DIPOLE SHIELDING PIPE

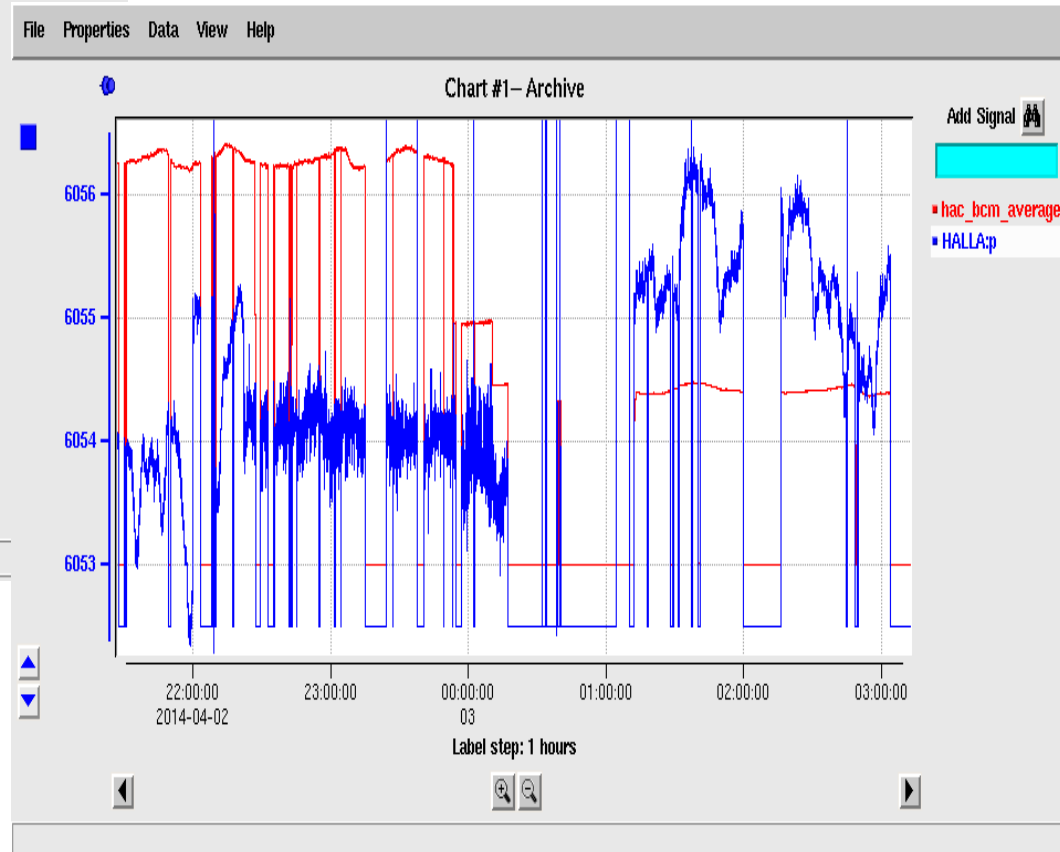
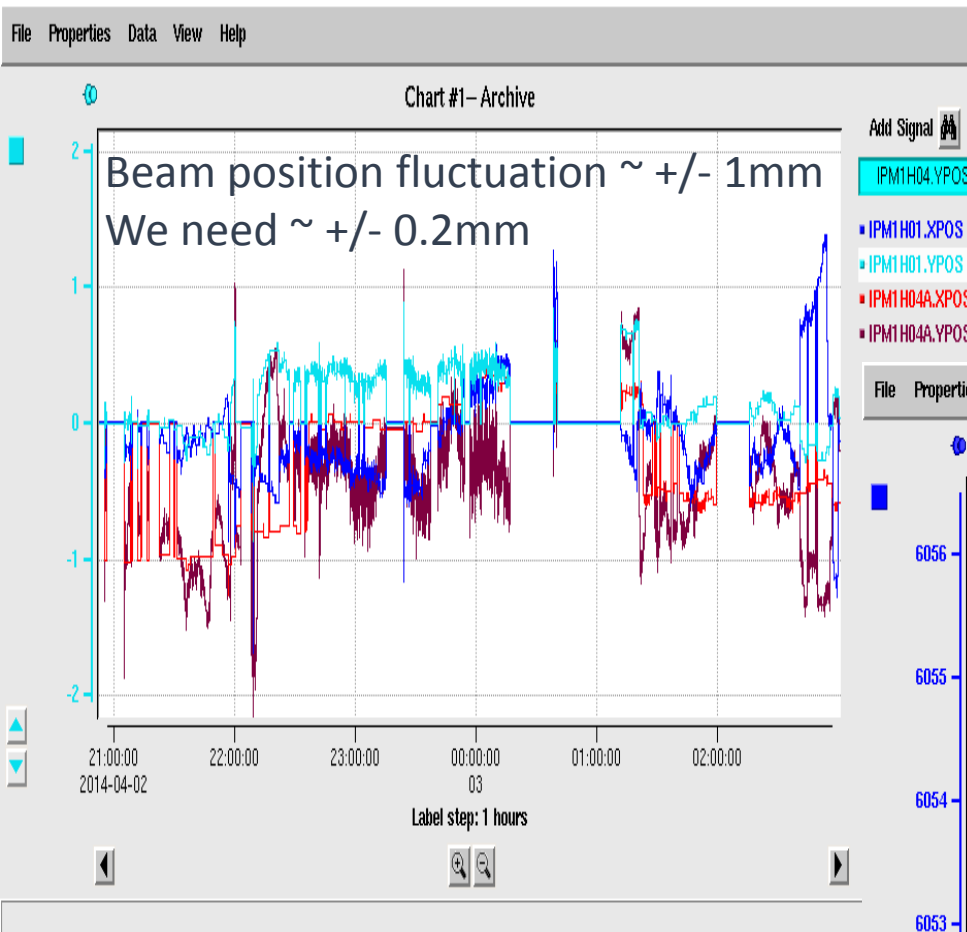


TOSCA simulation



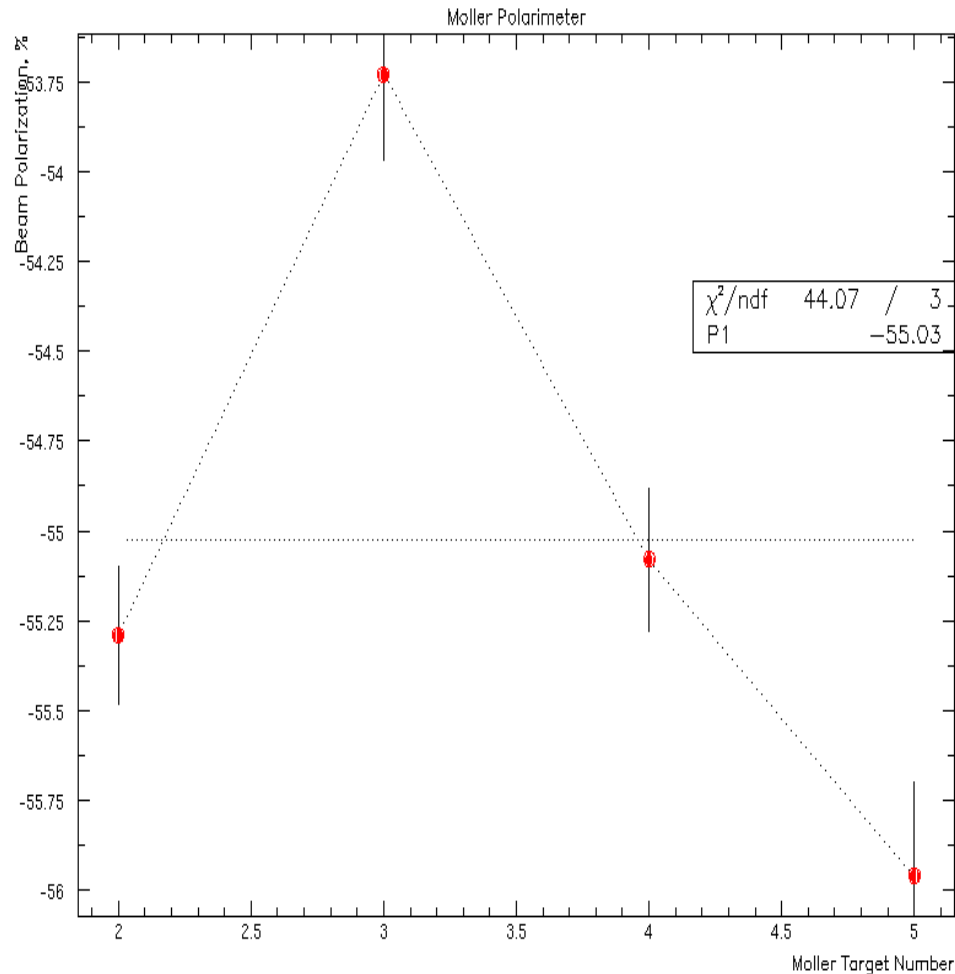
BEAM QUALITY

Can't study optics (quads scan) and detector position due to beam energy and beam position instability

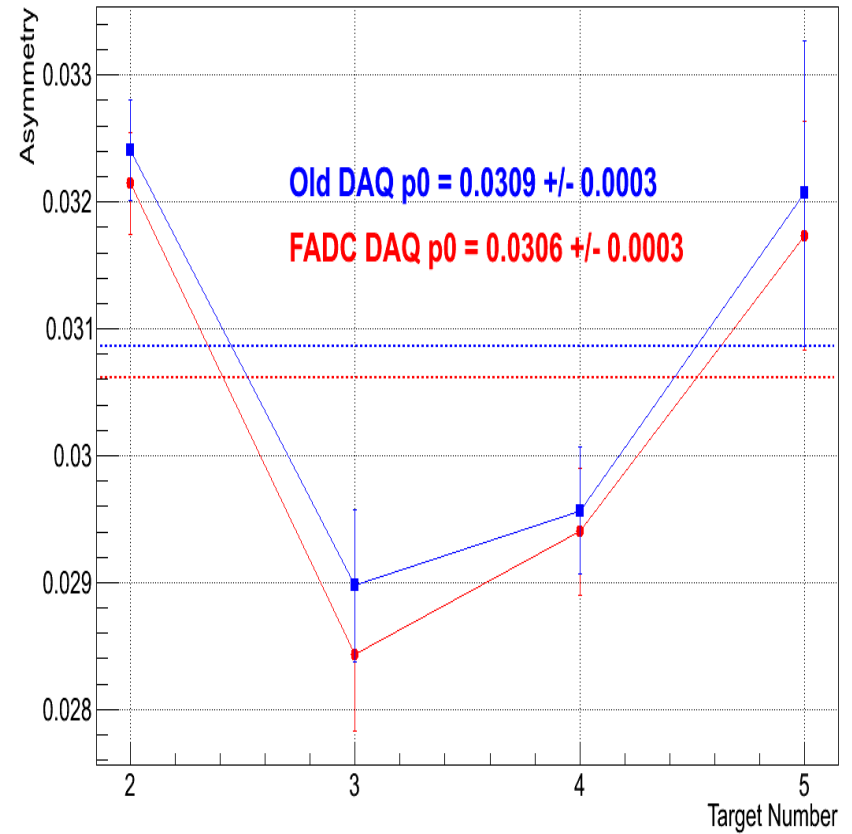


BEAM POLARIZATION MEASUREMENT

Moller result 04/02-03/2014

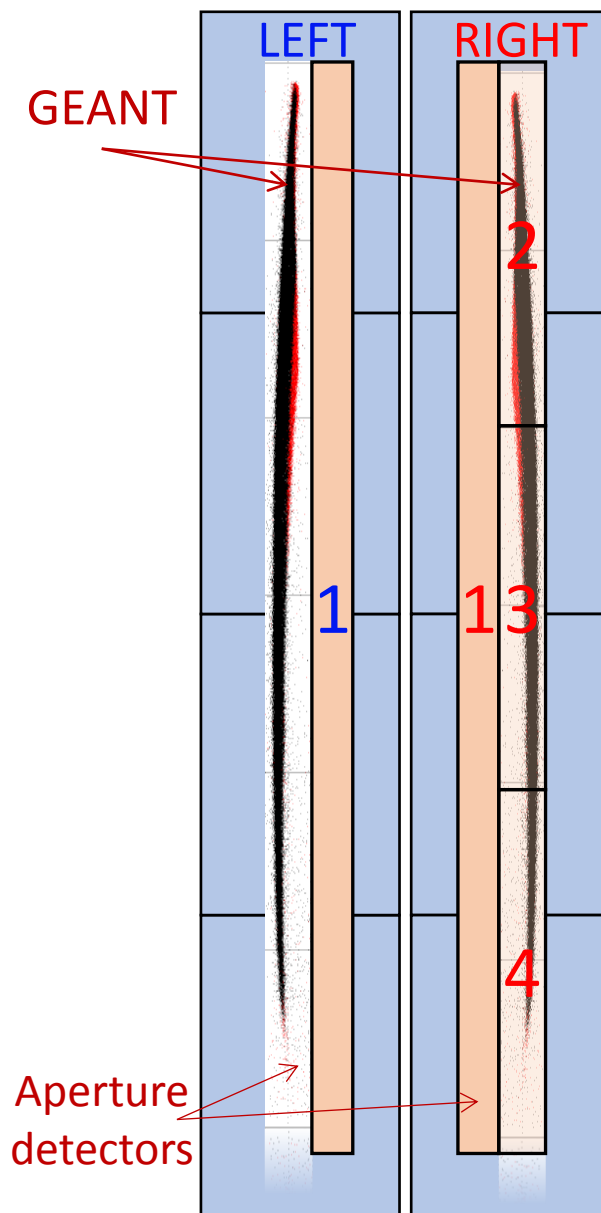
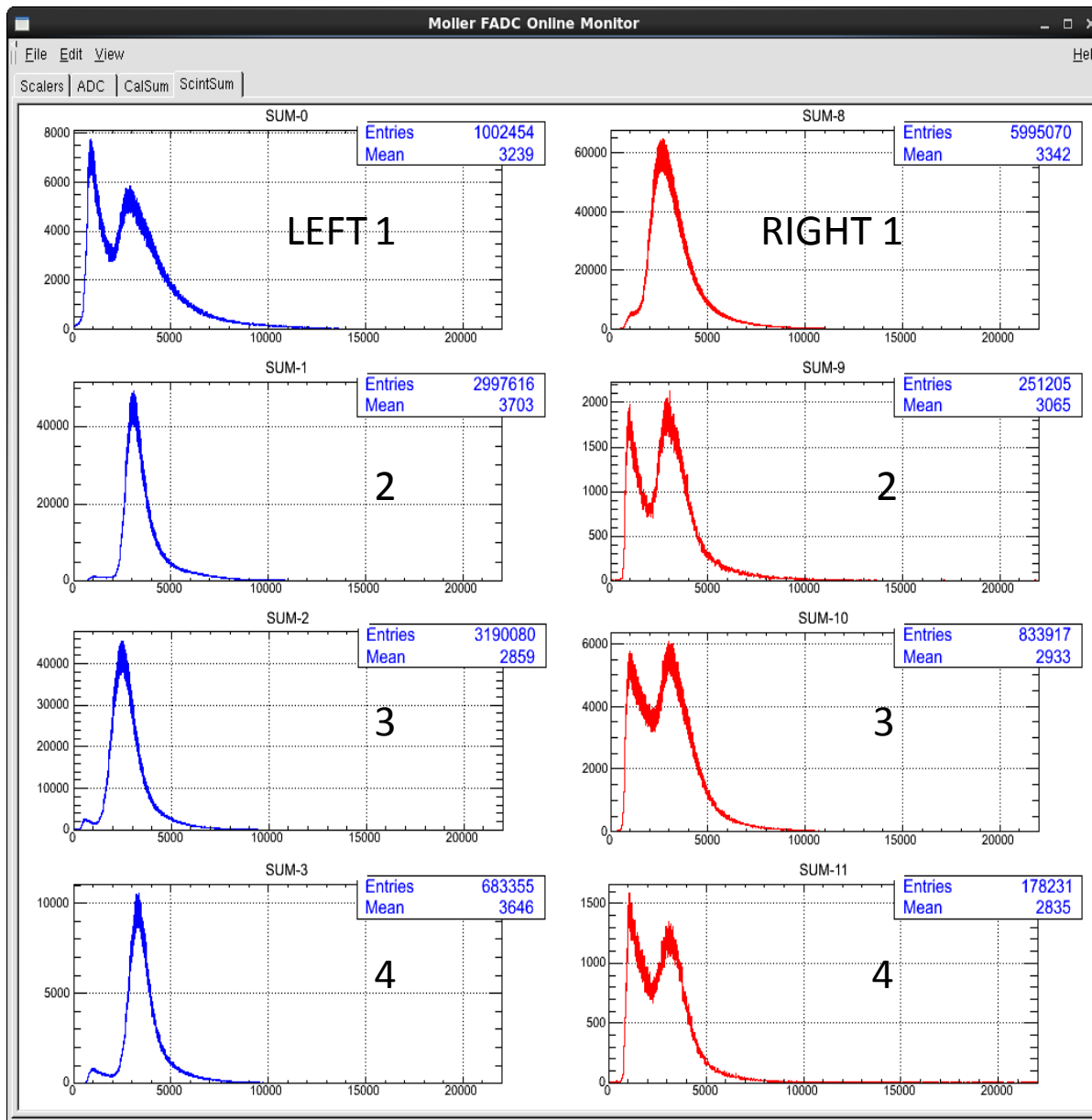


Asymmetry All Targets



Target Position	6	5	4	3	2	1	0
Material	park	SM	Fe	Fe	SM	SM	Al
Thickness, μm		6.8	9.3	14.3	29.4	13.0	16.5

DETECTOR MISALIGNMENT?



SUMMARY

1. Hall A Møller polarimeter is back to operation
2. Old DAQ and FADC DAQ are working
3. Detector misalignment has to be checked
4. More systematic studies have to be done

**Thanks to ALL for hard work on upgrading of
Hall A Møller polarimeter !**

Thank you for your attention!

Systematic Errors for Current System

Variable	"Low" field	"High" field 3.9T
<i>Target polarization</i>	1.5%	0.35%
<i>Analyzing power</i>	0.3%	0.3%
<i>Levchuk effect</i>	0.2%	0.3%
<i>Target temperature</i>	0.02%	0.02%
<i>Dead time</i>	0.3%	0.3%
<i>Background</i>	0.3%	0.3%
<i>High beam current</i>	0.2%	0.2%
<i>Others</i>	0.5%	0.5%
<i>Total</i>	1.7%	0.9%

Calorimeter Amplitude Spectrum

